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The optimal $\log\log q$ bound in the Pólya–Vinogradov inequality

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Abstract

Using the 11/12 consequence of OpenAI's formalised 7/8 zero-free theorem for nonprincipal Dirichlet L-functions, we derive the bound $C\sqrt{q}\log\log q$ for every primitive Dirichlet character modulo $q \geq 3$. The zero-free theorem is cited as an external input. We give the subsequent contour, prime-sum, major-arc and minor-arc arguments in detail and prove the required weak Montgomery–Vaughan exponential-sum estimate using Selberg's sieve. The same input yields $|L'/L(1,\chi)| \ll \log\log q$ and two-sided bounds for cyclotomic Euler–Kronecker constants. Explicit parameters and a general transfer argument show why any fixed uniform zero-free boundary below one suffices. No independent proof of the external zero-free theorem is claimed.

1. Introduction

Let χ be a primitive Dirichlet character modulo $q \geq 3$. The Pólya–Vinogradov inequality states that $\left|\sum_{n\leq x}\chi(n)\right| \ll \sqrt{q}\log q$ uniformly in x [3, Chapter 23].[1] Paley proved that the factor $\log q$ cannot be replaced by anything smaller than $\log\log q$: for infinitely many primitive quadratic characters the maximum over x exceeds $c\sqrt{q}\log\log q$ [12]. Montgomery and Vaughan [9] proved the upper bound $\sqrt{q}\log\log q$ under the generalised Riemann hypothesis. Without hypothesis, Granville and Soundararajan lowered the exponent

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[1] The proof expands χ in additive characters through the Gauss sum $\tau(\overline{\chi})$, whose modulus is $\sqrt{q}$ for primitive χ, and bounds the resulting geometric series trivially; Lemma 5.1 below is the same argument with the harmonic sums kept.

of $\log q$ for characters of fixed odd order [5].[2] We use the following precisely identified external theorem to obtain the bound of Montgomery and Vaughan.

THEOREM 1.1 (Consequence of OpenAI, [11, Theorem 1.1]). *For every nonprincipal Dirichlet character ξ, one has $L(s,\xi) \neq 0$ whenever $\operatorname{Re} s > 11/12$.*

External theorem and deduction of the stated boundary. We use the stronger 7/8 theorem of [11]. Its formal statement applies to every Dirichlet character, with only the principal pole at $s = 1$ excepted. Nonprincipality excludes that exception. Since $7/8 < 11/12$, the displayed conclusion follows. No primitivity assumption or additional Euler-factor argument is needed for this formal statement. The corresponding declaration is

`OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re`.

The source and its attribution are retained in the accompanying repository. We invoke the credited formal theorem; no new independent proof of it is claimed here. □

THEOREM 1.2 (Using the cited zero-free input). *There is an absolute constant C such that for every primitive character χ modulo $q \geq 3$ and every $x \geq 1$,*

$$\Big|\sum_{n\leq x} \chi(n)\Big| \leq C\sqrt{q}\log\log q.$$

By Paley's theorem the order of this upper bound is optimal. The deductions in this article begin after Theorem 1.1; the zero-free theorem is explicitly attributed to its source. An independent proof of that theorem by the present authors is not claimed. The formal declaration above is the input used throughout; the numerical weaker boundary is chosen only to keep the following explicit parameters. Appendix A gives the explicit constants at the boundary 11/12, and Appendix B proves the transfer for every fixed boundary strictly smaller than one. Appendix C proves the resulting smoothed twisted Möbius estimate, including its dependence on nonvanishing and the converse implication.

Montgomery and Vaughan use the Riemann hypothesis in one place: on the major arcs, for prime sums of the characters $\chi\psi$ twisted by additive characters, where ψ runs over the characters of a small modulus r. A power saving in these sums is enough there, and Theorem 1.1 provides one (Corollary 3.8). On the minor arcs they use their unconditional bound for exponential sums with multiplicative coefficients. A weaker form of it suffices, and we prove that

[2] For a primitive character of odd order g they obtain $\sqrt{q}(\log q)^{1-\delta_g/2+o(1)}$ with $\delta_g = 1 - \frac{g}{\pi}\sin\frac{\pi}{g}$, and under the generalised Riemann hypothesis the same exponent of $\log\log q$.

form in Section 4 with Selberg's sieve as the only arithmetic input. Figure 1 shows how the parts fit together.

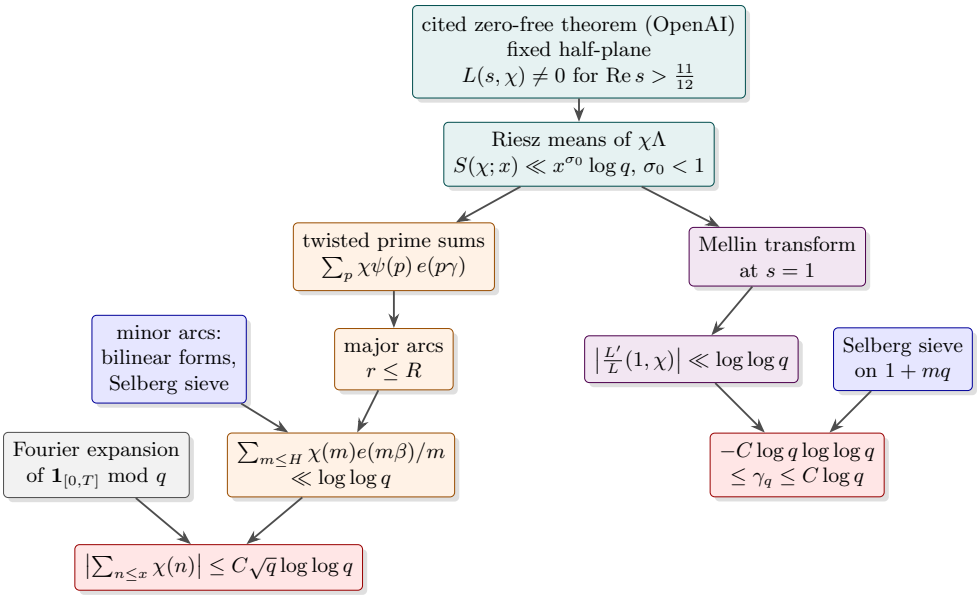


Figure 1. The structure of the proofs. The left branch proves Theorem 1.2: the Fourier expansion of a character sum reduces it to the sums $\sum_{m\le H}\chi(m)e(m\beta)/m$, which are bounded on the major arcs by the half-plane and on the minor arcs by bilinear forms and the sieve. The right branch proves Theorem 1.3.

The same half-plane controls the logarithmic derivative of $L(s,\chi)$ at $s=1$. For a number field K write $\zeta_K(s) = c_{-1}(s-1)^{-1} + c_0 + O(s-1)$ near $s=1$; the Euler–Kronecker constant of K is $\gamma_K = c_0/c_{-1}$. Let γ_q be the Euler–Kronecker constant of $\mathbb{Q}(\zeta_q)$. For a prime q the factorisation $\zeta_{\mathbb{Q}(\zeta_q)}(s) = \zeta(s)\prod_{\chi\neq\chi_0} L(s,\chi)$, the product over the nonprincipal characters modulo q [13, Theorem 4.3],[3] gives

$$\gamma_q = \gamma + \sum_{\chi\neq\chi_0} \frac{L'}{L}(1,\chi), \tag{1.1}$$

where γ is Euler's constant.

THEOREM 1.3. *Using Theorem 1.1, there is an absolute constant C with the following properties.*

[3]All nonprincipal characters modulo a prime are primitive, so no Euler factors have to be removed: the conductor of each nonprincipal χ is q itself, and $L(s,\chi)$ is the L-function of the corresponding primitive character.

(i) *For every nonprincipal character χ modulo q,*

$$\Big|\frac{L'}{L}(1,\chi)\Big| \le C\log\log q.$$

(ii) *For every prime $q \ge 3$, $-C\log q\log\log q \le \gamma_q \le C\log q$.*

Under the generalised Riemann hypothesis the bound (i) is classical, Ihara proved $\gamma_q \le (2+o(1))\log q$ and Badzyan proved a lower bound of the order of (ii) [7, 1]. Ford, Luca and Moree proved the two bounds of (ii) without hypothesis for a set of primes q of density one [4, 10].[4] For all primes the best unconditional bound we know is $|\gamma_q| \ll_\varepsilon q^\varepsilon$, due to Ihara, Murty and Shimura [8]. The upper bound in (ii) needs only Chebyshev's bound. The lower bound needs an upper bound for the prime powers $n \le q^{25}$ with $n \equiv 1 \pmod q$, which we obtain from Selberg's sieve applied in dyadic blocks (Lemma 6.3).

The arguments of Theorems 1.2 and 1.3 work, with other constants, whenever $L(s,\chi) \ne 0$ on $\operatorname{Re} s > \theta$ for all nonprincipal χ and some fixed $\theta < 1$. The paper is organised as follows. Section 2 collects Chebyshev's bounds and Selberg's sieve. Section 3 bounds Riesz means of $\chi\Lambda$ by a contour shift. Section 4 proves the bound for exponential sums with multiplicative coefficients, Section 5 proves Theorem 1.2 and Section 6 proves Theorem 1.3.

2. **Preliminaries**

Throughout, p and ℓ denote primes, Λ is the von Mangoldt function, $\theta(x) = \sum_{p\le x}\log p$, $\psi(x) = \sum_{n\le x}\Lambda(n)$, $e(x) = e^{2\pi i x}$, and $\|x\|$ is the distance from x to the nearest integer. We write $\omega(n)$ for the number of prime factors of n, $\tau_3(n)$ for the number of ordered factorisations $n = abc$, and $P(n)$ for the largest prime factor of $n \ge 2$, with $P(1) = 1$.

LEMMA 2.1. *For $x \ge 1$ and integers $N \ge 2$,*

(i) $\theta(x) \le x\log 4$;
(ii) $\psi(x) - \theta(x) \le 2\sqrt{x}\log x$ *and* $\psi(x) \le 4x$;
(iii) $\sum_{p\le N}\frac{\log p}{p} \le \log N + \log 4$;
(iv) $\sum_{p\le N}\frac{1}{p} \le \log\log N + 10$.

Proof. (i) It suffices to treat integers $m \ge 1$, and we use induction on m. The cases $m = 1, 2$ are clear, and for even $m > 2$ we have $\theta(m) = \theta(m-1)$. Let $m = 2n+1$. The primes in $(n+1, 2n+1]$ divide $\binom{2n+1}{n}$, and $2\binom{2n+1}{n} \le$

[4] The difficulty for an individual prime q lies in the prime powers $n \equiv 1 \pmod q$, which enter γ_q with the weight $(q-1)\Lambda(n)/n$. Below q^{25} we bound them for each prime q separately by Lemma 6.3; beyond q^{25} the zero-free half-plane makes their contribution bounded.

2^{2n+1} because $\binom{2n+1}{n}$ and $\binom{2n+1}{n+1}$ are equal terms of the binomial expansion of $(1+1)^{2n+1}$. Hence $\theta(2n+1) \le \theta(n+1) + n\log 4 \le (2n+1)\log 4$.

(ii) We have $\psi(x) - \theta(x) = \sum_{k\ge 2} \theta(x^{1/k})$, where only $k \le \log x/\log 2$ contribute. Each term is at most $\sqrt{x}\log 4$ by (i), so the sum is at most $(\log x/\log 2)\sqrt{x}\log 4 = 2\sqrt{x}\log x$. Since $\log x \le \sqrt{x}$, $\psi(x) \le (\log 4 + 2)x \le 4x$.

(iii) By Legendre's formula and (i),

$$N\log N \ge \log N! \ge \sum_{p\le N} \Big\lfloor \frac{N}{p} \Big\rfloor \log p \ge N \sum_{p\le N} \frac{\log p}{p} - N\log 4.$$

(iv) Let $A(u) = \sum_{p\le u} \log p/p$. Partial summation and (iii) give

$$\sum_{p\le N} \frac{1}{p} = \frac{A(N)}{\log N} + \int_2^N \frac{A(u)}{u\log^2 u}\,du \le 1 + \frac{\log 4}{\log 2} + \int_2^N \Big(\frac{1}{u\log u} + \frac{\log 4}{u\log^2 u}\Big)du,$$

and the integral is at most $\log\log N - \log\log 2 + 2$. □

We use Selberg's upper bound sieve in the following form; for the method see [6, Chapter 3], and we give the proof of the form we need.[5]

LEMMA 2.2. *Let $a_1, \dots, a_X$ be integers, P a squarefree integer, ν a multiplicative function with $0 < \nu(p) < 1$ for $p \mid P$, and $y \ge 1$. For $d \mid P$ put $R_d = \#\{n \le X : d \mid a_n\} - \nu(d)X$, and for $l \mid P$ put $g(l) = \prod_{p\mid l} \nu(p)/(1-\nu(p))$. Then*

$$\#\{n \le X : (a_n, P) = 1\} \le \frac{X}{S} + \sum_{\substack{d\mid P\\ d\le y}} 3^{\omega(d)}|R_d|, \qquad S = \sum_{\substack{l\mid P\\ l\le\sqrt{y}}} g(l).$$

Proof. Let D be the set of divisors d of P with $d \le \sqrt{y}$, and let $(\lambda_d)_{d\in D}$ be real numbers with $\lambda_1 = 1$. If $(a_n, P) = 1$ then $\sum_{d\in D,\, d\mid a_n} \lambda_d = 1$, and otherwise this sum is real, so

$$\begin{aligned}\#\{n : (a_n, P) = 1\} &\le \sum_{n\le X} \Big(\sum_{\substack{d\in D\\ d\mid a_n}} \lambda_d\Big)^2 \\ &= X \sum_{d_1,d_2\in D} \lambda_{d_1}\lambda_{d_2}\nu([d_1,d_2]) + \sum_{d_1,d_2\in D} \lambda_{d_1}\lambda_{d_2} R_{[d_1,d_2]}.\end{aligned}$$

Write Q for the double sum in the first term. Since ν is multiplicative and d_1, d_2 are squarefree, $\nu([d_1,d_2]) = \nu(d_1)\nu(d_2)/\nu((d_1,d_2))$. Let h be the multiplicative function on the divisors of P with $h(p) = 1/\nu(p) - 1 = 1/g(p)$; then $1/\nu(m) =$

[5] The weights λ_d are those of Selberg's quadratic form, and the only feature used later is the bound $|\lambda_d| \le 1$, which makes the remainder term at most a sum of $3^{\omega(d)}|R_d|$ over $d \le y$.

$\sum_{l\mid m} h(l)$ for $m \mid P$, and

$$Q = \sum_{l\mid P} h(l)\Big(\sum_{\substack{d\in D\\ l\mid d}} \lambda_d \nu(d)\Big)^2 = \sum_{l\in D} \frac{y_l^2}{g(l)}, \qquad y_l = \sum_{\substack{d\in D\\ l\mid d}} \lambda_d \nu(d).$$

Möbius inversion gives $\lambda_d \nu(d) = \sum_{l\in D,\, d\mid l} \mu(l/d) y_l$, so the condition $\lambda_1 = 1$ reads $\sum_{l\in D} \mu(l) y_l = 1$. By Cauchy's inequality

$$1 \le \Big(\sum_{l\in D} \frac{y_l^2}{g(l)}\Big)\Big(\sum_{l\in D} g(l)\Big) = QS,$$

with equality for $y_l = \mu(l) g(l)/S$. We make this choice, so that $Q = 1/S$. Then, writing $l = de$,

$$\lambda_d = \frac{1}{\nu(d) S} \sum_{\substack{l\in D\\ d\mid l}} \mu(l/d)\mu(l) g(l) = \frac{\mu(d) g(d)}{\nu(d) S} \sum_{\substack{e\in D,\ (e,d)=1\\ de\le \sqrt{y}}} g(e).$$

Since $g(d)/\nu(d) = \prod_{p\mid d}(1-\nu(p))^{-1} = \prod_{p\mid d}(1+g(p)) = \sum_{f\mid d} g(f)$, we get $|\lambda_d| \le S^{-1} \sum_{f\mid d} \sum_e g(fe)$, and the products fe are distinct elements of D; hence $|\lambda_d| \le 1$. Finally $[d_1, d_2] \le d_1 d_2 \le y$, and every $d \mid P$ is the least common multiple of at most $3^{\omega(d)}$ pairs (d_1, d_2), so the second term is at most $\sum_{d\mid P,\, d\le y} 3^{\omega(d)} |R_d|$. □

3. **Riesz means of prime sums**

For an arithmetic function a and $x \ge 1$ put

$$S(a; x) = \sum_{n\le x} a(n)\Big(1 - \frac{n}{x}\Big), \qquad S(\chi; x) = S(\chi\Lambda; x) = \sum_{n\le x} \chi(n)\Lambda(n)\Big(1 - \frac{n}{x}\Big).$$

The weight $1 - n/x$ makes the Mellin kernel absolutely integrable, which keeps the contour shift below free of truncation errors. The aim of this section is Proposition 3.7.

LEMMA 3.1. *Let $a\colon \mathbb{N} \to \mathbb{C}$ satisfy $\sum_n |a(n)| n^{-\sigma} < \infty$ for $\sigma > 1$, and let $F(s) = \sum_n a(n) n^{-s}$ for $\operatorname{Re} s > 1$.*

(i) *For $\operatorname{Re} s > 1$, $\displaystyle\int_1^\infty S(a; x)\, x^{-s-1}\, dx = \frac{F(s)}{s(s+1)}$.*

(ii) *For $x > 0$, $\displaystyle S(a; x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(2+it)\, \frac{x^{2+it}}{(2+it)(3+it)}\, dt$.*

In other words the following square commutes, and (ii) inverts its right column on the line $\operatorname{Re} s = 2$*:*

$$\begin{array}{ccc} a & \xrightarrow{\text{Riesz mean}} & S(a;\cdot) \\ \Big\downarrow{\scriptstyle \text{Dirichlet series}} & & \Big\downarrow{\scriptstyle \text{Mellin transform}} \\ F & \xrightarrow[\times\, \frac{1}{s(s+1)}]{} & \dfrac{F(s)}{s(s+1)} \end{array}$$

Proof. (i) For $n \geq 1$, $\int_n^\infty (1-n/x)x^{-s-1}dx = n^{-s}/s - n^{-s}/(s+1) = n^{-s}/(s(s+1))$. Since $\sum_n |a(n)| \int_n^\infty x^{-\sigma-1}dx = \sigma^{-1}\sum_n |a(n)|n^{-\sigma} < \infty$, the sum and the integral may be exchanged.

(ii) For $y > 0$ let $k(y) = \frac{1}{2\pi i}\int_{(2)} y^s \frac{ds}{s(s+1)}$. If $y \geq 1$, moving the line of integration to $\operatorname{Re} s = -U$, with $U > 2$, picks up the residues 1 at $s = 0$ and $-1/y$ at $s = -1$. For each fixed U, the horizontal integrals tend to zero as their height tends to infinity: their lengths are fixed and the denominator is of order the square of that height. On the new vertical line,

$$\int_{-\infty}^{\infty} \frac{|y^{-U+it}|}{|(-U+it)(1-U+it)|}\,dt \leq y^{-U}\int_{-\infty}^{\infty} \frac{dt}{(U-1)^2+t^2} = \frac{\pi y^{-U}}{U-1}.$$

This tends to zero for $y \geq 1$, including $y = 1$, so $k(y) = 1-1/y$ there. For $y < 1$, shifting instead to $\operatorname{Re} s = U$ crosses no pole and leaves an integral bounded by $\pi y^U/U$, which tends to zero. As $|s(s+1)|^{-1}$ is integrable on $\operatorname{Re} s = 2$ and $\sum_n |a(n)|n^{-2} < \infty$, summation and integration may be exchanged, and the right side equals $\sum_n a(n)k(x/n) = S(a;x)$. □

LEMMA 3.2. *Let* χ *be nonprincipal modulo* q*. For* $\operatorname{Re} s = \sigma > 0$*,* $|L(s,\chi)| \leq q|s|/\sigma$*.*

Proof. Put $A(t) = \sum_{n\leq t}\chi(n)$; then $|A(t)| \leq q$, and partial summation gives $L(s,\chi) = s\int_1^\infty A(t)t^{-s-1}\,dt$ for $\sigma > 0$. Hence $|L(s,\chi)| \leq q|s|\int_1^\infty t^{-\sigma-1}dt = q|s|/\sigma$. □

LEMMA 3.3. *Let* $A_0 = \frac{3}{2}\sum_\ell \ell^{-3/2}$*. For every Dirichlet character* χ *the series* $E(s) = -\sum_\ell \log(1-\chi(\ell)\ell^{-s})$ *is holomorphic on* $\operatorname{Re} s > 1$*,* $e^{E(s)} = L(s,\chi)$ *there, and* $|E(s)| \leq A_0$ *for* $\operatorname{Re} s \geq 3/2$*.*

Proof. For $\operatorname{Re} s \geq 1$ we have $|\chi(\ell)\ell^{-s}| \leq 1/2$, and $|\log(1-w)| \leq \sum_k |w|^k/k \leq \frac{3}{2}|w|$ for $|w| \leq 1/2$: indeed, the terms with $k \geq 2$ have total at most $|w|^2/(2(1-|w|)) \leq |w|/2$. The series therefore converges locally uniformly on $\operatorname{Re} s > 1$ and is bounded by A_0 on $\operatorname{Re} s \geq 3/2$, and the Euler product gives $e^E = L$. □

LEMMA 3.4 (Borel–Carathéodory). *Let h be holomorphic on a neighbourhood of the disc $|z-c|\le R$ and let $0<r<R$. Then*

$$\max_{|z-c|\le r}|h(z)-h(c)|\le\frac{2r}{R-r}\max_{|w-c|=R}\operatorname{Re}\bigl(h(w)-h(c)\bigr).$$

Proof. Put $g(z)=h(c+z)-h(c)$ and $A=\max_{|w|=R}\operatorname{Re}g(w)$. The mean of $\operatorname{Re}g$ over the circle $|w|=R$ is $\operatorname{Re}g(0)=0$, so $A\ge0$. If $A=0$, then $\operatorname{Re}g$ vanishes on the circle, hence on the disc, and g is an imaginary constant equal to $g(0)=0$. Let $A>0$. By the maximum principle $\operatorname{Re}g\le A$ on $|z|\le R$, so $|2A-g|\ge A$ there and $\phi=g/(2A-g)$ is holomorphic. Since $|g|^2-|2A-g|^2=4A(\operatorname{Re}g-A)\le0$, we have $|\phi|\le1$, and $\phi(0)=0$. Schwarz's lemma gives $|\phi(z)|\le|z|/R$, and from $g=2A\phi/(1+\phi)$ we get $|g(z)|\le2A|z|/(R-|z|)$. □

LEMMA 3.5. *Let $0<\delta<1/1000$, $1/2\le a\le1$, and let χ be nonprincipal modulo q with $L(s,\chi)\ne0$ for $\operatorname{Re}s>a+2\delta$. Put $K_1=(1+2A_0)/(2\delta^2)$. For $a+8\delta\le\sigma\le2$ and $t\in\mathbb{R}$,*

$$\Bigl|\frac{L'}{L}(\sigma+it,\chi)\Bigr|\le K_1\log\bigl(2q(3+|t|)^2\bigr).$$

Proof. Let $c=2+it$, $R=2-a-3\delta$, $r=2-a-6\delta$ and $\Lambda_0=\log(2q(3+|t|)^2)\ge\log18>1$. On the disc $|z-c|<2-a-2\delta$ we have $\operatorname{Re}z>a+2\delta$, so $L(z,\chi)$ has no zero there and has a holomorphic logarithm h. On $|z-c|<1/2$ both h and the function E of Lemma 3.3 are logarithms of L, so they differ by a constant in $2\pi i\mathbb{Z}$; we choose $h=E$ there. Then $|h(c)|\le A_0$.

If $|z-c|\le R$, then $\operatorname{Re}z\ge a+3\delta\ge1/2$ and $|z|\le|c|+3/2\le4+|t|$, so Lemma 3.2 gives $\operatorname{Re}h(z)=\log|L(z,\chi)|\le\log(2q(4+|t|))\le\Lambda_0$. Lemma 3.4 gives, for $|z-c|\le r$,

$$|h(z)-h(c)|\le\frac{2r}{R-r}(\Lambda_0+A_0)\le\frac{\Lambda_0+A_0}{\delta},$$

as $r\le3/2$ and $R-r=3\delta$. Hence $|h(z)|\le A_0+(\Lambda_0+A_0)/\delta\le(1+2A_0)\Lambda_0/\delta$ on $|z-c|\le r$. For $w=\sigma+it$ with $a+8\delta\le\sigma\le2$ we have $|w-c|=2-\sigma\le r-2\delta$, so the circle $|z-w|=2\delta$ lies in $|z-c|\le r$, and Cauchy's estimate gives $|L'/L(w,\chi)|=|h'(w)|\le(1+2A_0)\Lambda_0/(2\delta^2)$. □

LEMMA 3.6. *For $q\ge1$, $t\in\mathbb{R}$ and $s=\sigma+it$ with $\sigma\ge1/2$,*

$$\log\bigl(2q(3+|t|)^2\bigr)\le10\log(q+2)(1+|t|)^{1/2},\qquad|s(s+1)|\ge\tfrac18(1+|t|)^2.$$

Proof. We have $\log(2q)\le2\log(q+2)$, and $\log y\le2y^{1/2}$ with $3+|t|\le4(1+|t|)$ gives $\log(3+|t|)\le4(1+|t|)^{1/2}$; the first bound follows from $\log(q+2)\ge1$. For the second, $|s|\ge\max(\sigma,|t|)\ge(1+|t|)/4$ and $|s+1|\ge\max(\sigma+1,|t|)\ge(1+|t|)/2$. □

PROPOSITION 3.7. *Let $0 < \delta < 1/1000$. There is $K = K(\delta)$ such that the following holds. Let $1/2 \le a \le 1$ and let χ be nonprincipal modulo q with $L(s,\chi) \neq 0$ for $\operatorname{Re} s > a + 2\delta$. Then for $x \ge 1$,*

$$|S(\chi;x)| \le K\, x^{a+8\delta} \log(q+2).$$

Proof. Put $\sigma_0 = a + 8\delta$ and

$$\Phi(s) = -\frac{L'}{L}(s,\chi)\,\frac{x^s}{s(s+1)}.$$

For $\operatorname{Re} s > 1$, $-L'/L(s,\chi) = \sum_n \chi(n)\Lambda(n)n^{-s}$, so Lemma 3.1(ii) gives $S(\chi;x) = \frac{1}{2\pi}\int \Phi(2+it)\,dt$. The function Φ is holomorphic on $\sigma_0 \le \operatorname{Re} s \le 2$, and by Lemmas 3.5 and 3.6,

$$|\Phi(\sigma+it)| \le 80K_1 \log(q+2)\, x^{\sigma}(1+|t|)^{-3/2} \qquad (\sigma_0 \le \sigma \le 2).$$

This tends to 0 as $|t| \to \infty$ uniformly in σ, so Cauchy's theorem on the rectangles $[\sigma_0, 2] \times [-T, T]$, $T \to \infty$, moves the integral to $\operatorname{Re} s = \sigma_0$ (Figure 2). Since $\int_{\mathbb{R}} (1+|t|)^{-3/2} dt = 4$, we get $|S(\chi;x)| \le \frac{160K_1}{\pi} x^{\sigma_0} \log(q+2)$. □

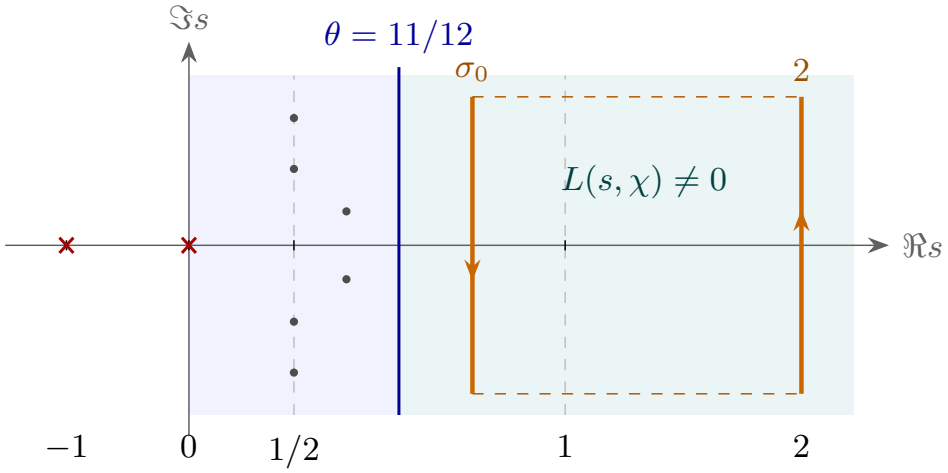


Figure 2. The contour shift in the proof of Proposition 3.7. The indicated nontrivial zeros of $L(s,\chi)$ lie in $0 \le \operatorname{Re} s \le 11/12$ (dots), the kernel $x^s/(s(s+1))$ has poles at 0 and -1 (crosses), and the line of integration moves from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = \sigma_0$ inside the zero-free half-plane. Horizontal spacing is schematic.

From now on fix $\delta = 1/2000$ and $a = 11/12 - 1/1000$, so that $a + 2\delta = 11/12$ and

$$\sigma_0 = a + 8\delta = \frac{11}{12} + \frac{3}{1000},$$

and let K be the constant of Proposition 3.7 for this δ. By Theorem 1.1, Proposition 3.7 applies to every nonprincipal character.

COROLLARY 3.8. *There is an absolute constant C_3 such that for every nonprincipal character ξ modulo k:*

(i) $|S(\xi;x)| \le Kx^{\sigma_0}\log(k+2)$ *for* $x \ge 1$;
(ii) $\left|\sum_{p\le x}\xi(p)\log p\right| \le C_3 x^{49/50}\log(k+2)$ *for* $x \ge 1$;
(iii) *for integers* $2 \le A \le B$ *and* $\gamma \in \mathbb{R}$,

$$\Big|\sum_{A<p\le B}\xi(p)e(p\gamma)\Big| \le 2C_3\,\frac{B^{49/50}\log(k+2)(2+2\pi|\gamma|B)}{\log A}. \tag{3.1}$$

Proof. (i) is Proposition 3.7. For (ii) put $T(y) = yS(\xi;y)$, that is, $T(y) = \sum_{n\le y}\xi(n)\Lambda(n)(y-n)$, and let $0 < h \le x$. Then

$$T(x+h) - T(x) = h\sum_{n\le x}\xi(n)\Lambda(n) + \sum_{x<n\le x+h}\xi(n)\Lambda(n)(x+h-n),$$

and the last sum has at most $h+1$ terms, each of modulus at most $h\log(2x)$. With $|T(y)| \le Ky^{1+\sigma_0}\log(k+2)$ this gives

$$\Big|\sum_{n\le x}\xi(n)\Lambda(n)\Big| \le \frac{2K(2x)^{1+\sigma_0}\log(k+2)}{h} + (h+1)\log(2x).$$

Take $h = x^{47/50}$. Since $1+\sigma_0-47/50 < 49/50$ and $\log(2x) \le 1+100x^{1/100}$, the right side is at most $(8K+202)x^{49/50}\log(k+2)$. The prime powers contribute at most $\psi(x)-\theta(x) \le 2\sqrt{x}\log x \le 200x^{49/50}$ by Lemma 2.1(ii), and (ii) follows with $C_3 = 8K+402$.

(iii) Let $\Pi(u) = \sum_{p\le u}\xi(p)\log p$ and $f(u) = e(u\gamma)/\log u$. Partial summation gives

$$\sum_{A<p\le B}\xi(p)e(p\gamma) = f(B)\Pi(B) - f(A)\Pi(A) - \int_A^B \Pi(u)f'(u)\,du,$$

and $|f'(u)| \le 2\pi|\gamma|/\log A + 1/(u\log^2 u)$ for $u \ge A$. By (ii) the right side is at most $C_3B^{49/50}\log(k+2)\big(3+2\pi|\gamma|B\big)/\log A$, which gives (3.1). □

4. **Exponential sums with multiplicative coefficients**

For $M > 0$ put $w_M(\theta) = M$ if $\|\theta\| < 1/M$ and $w_M(\theta) = 1/\|\theta\|$ otherwise; then $0 \le w_M(\theta) \le M$ and $w_M(-\theta) = w_M(\theta)$.

THEOREM 4.1. *There is an absolute constant C_1 with the following property. Let f be completely multiplicative with $|f(n)| \le 1$, let $\alpha \in \mathbb{R}$, and let a, r*

be coprime integers with $r \ge 1$, $|\alpha - a/r| \le r^{-2}$ *and* $2 \le R \le r \le N/R$. *Then*

$$\Big|\sum_{n\le N} f(n)e(n\alpha)\Big| \le C_1\Big(\frac{N}{\log N} + \frac{N(\log N)^2}{R^{1/4}}\Big).$$

Montgomery and Vaughan [9, Corollary 1] proved this for multiplicative f with $N(\log R)^{3/2}R^{-1/2}$ in place of $N(\log N)^2R^{-1/4}$. The proof below follows their approach, a bilinear decomposition together with the spacing of the multiples of α and an upper bound for prime pairs, with cruder estimates, which suffice for Theorem 1.2.

LEMMA 4.2. *Let* a, r *be coprime with* $r \ge 2$ *and* $|\alpha - a/r| \le r^{-2}$, *and let* $H \ge 0$ *be an integer.*

(i) $\|k\alpha\| \ge 1/(2r)$ *for every integer* k *with* $0 < |k| \le r/2$.

(ii) $\#\{1 \le h \le H : \|h\alpha\| < t\} \le (4H/r + 1)(4rt + 1)$ *for every* $t \ge 0$.

(iii) *For* $M > 0$ *and every integer* $K \ge 0$ *with* $M \le 2^K$,

$$\sum_{h\le H} w_M(h\alpha) \le \Big(\frac{4H}{r} + 1\Big)(3M + 4r + 8rK),$$

$$\sum_{h\le H} w_M(h\alpha)^2 \le \Big(\frac{4H}{r} + 1\Big)(20rM + 3M^2).$$

Proof. (i) Let z be the integer nearest to $k\alpha$. Since $0 < |k| < r$ and $(a, r) = 1$, $r \nmid ka$, so $|ka/r - z| \ge 1/r$ and $\|k\alpha\| \ge 1/r - |k|\,|\alpha - a/r| \ge 1/(2r)$.

(ii) Let $L = \lfloor r/2 \rfloor$, so $L \ge r/4$. For k in a block of L consecutive integers let $y_k = k\alpha - z_k$ with z_k the integer nearest to $k\alpha$. For $k \ne k'$ in the block, $0 < |k - k'| < L \le r/2$ and $y_k - y_{k'} \equiv (k - k')\alpha \pmod 1$, so $|y_k - y_{k'}| \ge 1/(2r)$ by (i). Points of $(-t, t)$ at mutual distances at least $1/(2r)$ number at most $4rt + 1$, and $[1, H]$ is covered by at most $H/L + 1 \le 4H/r + 1$ blocks.

(iii) If $\|\theta\| \ge 1/M$, let $k \ge 1$ be least with $\|\theta\| < 2^k/M$; then $k \le K$ because $\|\theta\| \le 1/2 < 2^K/M$, and $\|\theta\| \ge 2^{k-1}/M$, so $w_M(\theta) \le 2^{1-k}M$. Hence

$$w_M(\theta)^j \le M^j\,[\|\theta\| < 1/M] + \sum_{k=1}^{K}(2^{1-k}M)^j\,[\|\theta\| < 2^k/M] \qquad (j = 1, 2).$$

Summing over $h \le H$ with (ii), the first sum is at most $(4H/r + 1)\big(4r + M + \sum_{k\le K}(8r + 2^{1-k}M)\big)$ and the second at most $(4H/r + 1)\big(4rM + M^2 + \sum_{k\ge1}(16rM2^{-k} + 4M^24^{-k})\big)$. □

LEMMA 4.3. *Let* $\theta, \alpha \in \mathbb{R}$, $U > 0$, *and let* A, B, N *be nonnegative integers with* $B \le A + U$.

(i) $\big|\sum_{A<x\le B} e(x\theta)\big| \le w_U(\theta)$.

(ii) *For a finite set V of positive integers and $b : V \to \mathbb{C}$,*

$$\sum_{A<u\le B}\Big|\sum_{\substack{v\in V\\ uv\le N}} b(v)e(uv\alpha)\Big|^2 \le \sum_{v,v'\in V} |b(v)|\,|b(v')|\,w_U\big((v-v')\alpha\big).$$

(iii) *For $M > 0$ and an integer $H \ge 0$, $\sum_{1\le m,m'\le H} w_M((m-m')\alpha) \le H\big(M + 2\sum_{h\le H} w_M(h\alpha)\big)$.*

Proof. (i) If $\|\theta\| < 1/U$, the sum has at most U terms of modulus 1. Otherwise $\theta \notin \mathbb{Z}$, the sum has modulus at most $|e((B-A)\theta)-1|/|e(\theta)-1|$, and $|e(\theta)-1| = 2|\sin\pi\theta| \ge 4\|\theta\|$, so the modulus is at most $1/(2\|\theta\|)$. (ii) Expanding the square gives $\sum_{v,v'} b(v)\overline{b(v')}\sum_u e(u(v-v')\alpha)$, where u runs over the integers of $(A, \min(B, N/v, N/v')]$, an interval of length at most U; apply (i). (iii) For fixed m, the term $m' = m$ is $w_M(0) = M$, and each $h = |m-m'| \in [1,H]$ arises from at most two m'. □

Lemma 4.4. *Let $C_5 = 1 + 4\zeta(3/2)$. For every completely multiplicative f with $|f(n)| \le 1$, every $\alpha \in \mathbb{R}$ and every integer $N \ge 1$, the sum $S = \sum_{n\le N} f(n)e(n\alpha)$ satisfies*

$$\Big|S\log N - \sum_{p\le N}\log p\, f(p)\sum_{m\le N/p} f(m)e(pm\alpha)\Big| \le C_5 N.$$

Proof. Write $S\log N = S_1 + S_2$ with $S_1 = \sum_{n\le N} f(n)e(n\alpha)\log n$ and $S_2 = \sum_{n\le N} f(n)e(n\alpha)\log(N/n)$. Then $|S_2| \le N\log N - \log N! \le N$, since $N^N/N! \le e^N$. In S_1 write $\log n = \sum_{d|n}\Lambda(d)$ and $n = dm$; complete multiplicativity gives $\sum_{d\le N}\Lambda(d)f(d)\sum_{m\le N/d} f(m)e(dm\alpha)$. The terms with d prime form the sum in the statement, and the others have modulus at most $N\sum_p\sum_{k\ge 2}\log p/p^k \le 2N\sum_p \log p/p^2 \le 4\zeta(3/2)N$, since $\log p \le 2\sqrt{p}$. □

Let $G(h) = \prod_{p|h} p/(p-1)$.

Lemma 4.5. (i) *For every integer $X \ge 2$ and every even $h \ge 2$,*

$$\#\{0 \le n < X : n \text{ and } n+h \text{ prime}\} \le \frac{400\,G(h)^2 X}{(\log X)^2} + 3X^{3/5}.$$

(ii) *$\sum_{h\le X} G(h)^4 \le e^{30}X$ for every $X \ge 1$.*

Proof. (i) Apply Lemma 2.2 to $a_n = n(n+h)$, $0 \le n < X$, with $y = X^{1/5}$, $z = \lfloor y\rfloor$, P the product of the primes $p \le z$ with $p \nmid h$, and $\nu(d) = 2^{\omega(d)}/d$. Every $p \mid P$ is odd, so $\nu(p) = 2/p < 1$. For $d \mid P$ the residues n with $d \mid n(n+h)$ form $2^{\omega(d)}$ classes modulo d (namely $n \equiv 0$ or $-h \pmod p$ for each $p \mid d$, distinct as $p \nmid h$), and each class contains between $X/d-1$ and $X/d+1$ of the integers $0 \le n < X$; hence $|R_d| \le 2^{\omega(d)}$. Since the prime factors of d are at least 3, $3^{\omega(d)}|R_d| \le 6^{\omega(d)} \le d^2 \le y^2$, and the error term is at most $y^3 = X^{3/5}$.

Let $\mathcal{M}$ be the set of $m \le \sqrt{y}$ whose prime factors divide P. Since $\nu(p)/(1-\nu(p)) = \sum_{k\ge1}\nu(p)^k$ and $\nu(m) = 2^{\Omega(m)}/m$ for these m, $g(l)$ is the sum of $2^{\Omega(m)}/m$ over the m whose set of prime factors is that of l. An $m \in \mathcal{M}$ is counted in $g(l)$ for l the product of its prime factors, and $l \le m \le \sqrt{y}$. Hence, with $\tau(m) \le 2^{\Omega(m)}$ the number of divisors of m,

$$S \ge \sum_{m\in\mathcal{M}} \frac{2^{\Omega(m)}}{m} \ge \sum_{m\in\mathcal{M}} \frac{\tau(m)}{m}.$$

Let $T = \lfloor y^{1/4} \rfloor$ and let B be the set of $b \le T$ that are prime to h; these b are odd. For $b, b' \in B$ the product bb' lies in $\mathcal{M}$, and a given m is a product bb' in at most $\tau(m)$ ways, so $S \ge \big(\sum_{b\in B} 1/b\big)^2$. If E is a set of primes and $p \notin E$, every $b \le T$ prime to E is uniquely $p^k b'$ with b' prime to $E \cup \{p\}$, so the sum of $1/b$ over the $b \le T$ prime to E is at most $p/(p-1)$ times the same sum for $E \cup \{p\}$. Adding the prime factors of h one at a time gives

$$\sum_{b\in B} \frac{1}{b} \ge \frac{1}{G(h)} \sum_{b\le T} \frac{1}{b} \ge \frac{\log(T+1)}{G(h)}.$$

As $T + 1 > y^{1/4} = X^{1/20}$, we get $S \ge (\log X)^2/(400G(h)^2)$, that is, $X/S \le 400G(h)^2 X/(\log X)^2$. Finally, if $n < X$ and both n and $n + h$ are prime, then $n \le z$ or $(n(n + h), P) = 1$, so the count is at most $z + 1 + X/S + y^3 \le 400G(h)^2X/(\log X)^2 + 3X^{3/5}$.

(ii) For $t = p-1 \ge 1$, $(t+1)^5 \le (t+31)t^4$, that is, $(p/(p-1))^4 \le 1+30/p$. Hence $G(h)^4 \le \sum_d 30^{\omega(d)}/d$ over the squarefree $d \mid h$, and

$$\begin{aligned}\sum_{h\le X} G(h)^4 &\le \sum_{d\le X} \mu(d)^2 \frac{30^{\omega(d)}}{d} \cdot \frac{X}{d} \le X \prod_p \Big(1 + \frac{30}{p^2}\Big) \\ &\le X \exp\Big(30 \sum_{n\ge2} \frac{1}{n^2}\Big) \le e^{30} X. \qquad \square\end{aligned}$$

Proof of Theorem 4.1. Let S be the sum in the theorem, $L = \log N$ and $\ell = \lfloor \log_2 N \rfloor$. Since $N \ge rR \ge 4$, we have $L \ge 1$. If $\ell < 64$, then $L < 64 \log 2 < 45$ and $|S| \le N \le 45N/L$. Let now $\ell \ge 64$, and put $I = \lfloor \log_2 \ell \rfloor + 1$ and $J = \ell + 1 - 4I$. Then $16^I > \ell^4$, $1 \le J \le \ell \le 2L$, $I \le \ell \le 2L$, $2^I \le 2\ell$, $N < 2^{\ell+1} = 2^J 16^I$ and $J^2 2^J \le (\ell+1)^2 2^{\ell+1} \ell^{-4} \le 2N$. By Lemma 4.4, $|S|L \le C_5 N + |T_1| + |T_2|$, where T_1 and T_2 are the parts of $\sum_p \log p\, f(p) \sum_{m\le N/p} f(m) e(pm\alpha)$ with $p \le 2^J$ and with $p > 2^J$.

The small primes. Let $0 \le j < J$, $U = 2^j$, $H = \lfloor N/U \rfloor$, and for $U < x \le 2U$ let $B(x) = \sum_{m\le H,\, xm\le N} f(m) e(xm\alpha)$; the inner sum for a prime $p \in (U, 2U]$ is $B(p)$. By Cauchy's inequality, Lemma 4.3(ii), (iii) and Lemma 4.2(iii)

with $M = U$, $K = j$,

$$\sum_{U<p\le 2U} \log p\,|B(p)| \le \big(U\log^2(2U)\big)^{1/2} Q_j^{1/2},$$

where $Q_j = H\big(U + 2(4H/r+1)(3U+4r+8rj)\big)$. Since $HU \le N$, $UQ_j \le 7NU + (8+16j)Nr + 24N^2/r + (32+64j)N^2/U$, and $\log(2U) \le j+1$. Summing over $j < J$ with $j+1 \le J$, $\sum_{j<J} 2^{j/2} \le 3\cdot 2^{J/2}$, $\sqrt{(32+64j)2^{-j}} \le 8(j+1)(3/4)^j$ and $\sum_{j\ge 0}(j+1)^2(3/4)^j = 112$,

$$|T_1| \le 3\sqrt{7}J\sqrt{N2^J} + J^2 N\sqrt{24/r} + 896N + J^2\sqrt{24JNr}.$$

Now $J^2 2^J \le 2N$ gives $3\sqrt{7}J\sqrt{N2^J} \le 12N$; $r \ge R \ge \sqrt{R}$ gives $\sqrt{24/r} \le 5R^{-1/4}$; and $r\sqrt{R} \le rR \le N$ gives $\sqrt{24JNr} \le 5JNR^{-1/4}$. With $J \le 2L$ and $L \ge 1$, $|T_1| \le 908N + 60L^3 NR^{-1/4}$.

The large primes. For $p > 2^J$ we have $N/p < 16^I$, so the m in T_2 lie in $(0, 16^I - 1]$, the union of the blocks $(16^i - 1, 16^{i+1} - 1]$ with $0 \le i < I$. Fix i, let $A = 16^i - 1$, $U = 15\cdot 16^i$ and $X = \lfloor N/16^i \rfloor$, and assume $X \ge 1$, the block being empty otherwise. Only the primes $p \in (2^J, X]$ contribute, since $p \ge X+1$ and $m \ge 16^i$ give $pm > N$. Let V be the set of these primes, which are odd, and $B(m) = \sum_{p\in V,\, pm\le N} \log p\, f(p) e(pm\alpha)$. The block is $\sum_{A<m\le A+U} f(m)B(m)$, and by Cauchy's inequality, Lemma 4.3(ii) and $\log p \le \lambda = \log(X+1)$, its square is at most $U\lambda^2 W$, where

$$W = \sum_{p,p'\in V} w_U((p-p')\alpha) = |V|U + 2\sum_{h=1}^{X} w_U(h\alpha)\,\#\{p \in V : p + h \in V\}.$$

The count vanishes for odd h, and for even h Lemma 4.5(i), applied to $X+1$, bounds it by $400G(h)^2(X+1)\lambda^{-2} + 3(X+1)^{3/5}$. By Cauchy's inequality, Lemma 4.5(ii) and Lemma 4.2(iii), $\sum_{h\le X} G(h)^2 w_U(h\alpha) \le e^{15}\sqrt{X}Q^{1/2}$ with $Q = (4X/r+1)(20rU + 3U^2)$, and $\sum_{h\le X} w_U(h\alpha) \le XU$. Hence $U\lambda^2 W \le A_1 + 800e^{15}Z + A_3$, where

$$A_1 = U^2\lambda^2 X, \qquad Z = U(X+1)\sqrt{X}Q^{1/2}, \qquad A_3 = 6U^2\lambda^2(X+1)^{3/5}X.$$

Let $\kappa = \log(N+1) \ge \lambda$ and $T_i = 4^i\kappa\,(N(2N)^{3/5})^{1/2}$. Since $U^2X = 225\cdot 16^i(16^iX) \le 225\cdot 16^i N$ and $X + 1 \le 2N$, we get $A_1 \le 225T_i^2$ and $A_3 \le 1350T_i^2 \le 37^2 T_i^2$. With $w = UX \le 15N$, $X \le N/16^i$ and $R \le r \le N/R$,

$$\begin{aligned} Z^2 \le 4U^2X^3Q &= 320w^3X + \frac{48w^4}{r} + 80rw^3 + 12Uw^3 \\ &\le \frac{320\cdot 15^3N^4}{16^i} + \frac{48\cdot 15^4N^4}{R} + \frac{80\cdot 15^3N^4}{R} \\ &\quad + 180\cdot 15^3\cdot 16^i N^3, \end{aligned}$$

and every coefficient is at most 40^4, so $Z^{1/2} \le 40(N2^{-i} + 2NR^{-1/4} + 2^iN^{3/4})$. The block is therefore at most

$$\begin{aligned} &15T_i + 40\sqrt{800e^{15}}\big(N2^{-i} + 2NR^{-1/4} + 2^iN^{3/4}\big) + 37T_i \\ &\qquad \le K_b\big(N2^{-i} + NR^{-1/4} + 2^iN^{3/4} + T_i\big), \end{aligned}$$

with $K_b = 80\sqrt{800e^{15}} + 52$. Summing over $i < I$,

$$|T_2| \le K_b\big(2N + INR^{-1/4} + 2^IN^{3/4} + 4^I\kappa(N(2N)^{3/5})^{1/2}\big).$$

Here $I \le 2L^3$; $2^I \le 4L \le 16N^{1/4}$; $4^I \le 16L^2$ and $\kappa \le 2L$; and $\log x \le 15x^{1/15}$ gives $L^3(N(2N)^{3/5})^{1/2} \le 2 \cdot 15^3 N$. Hence $|T_2| \le K_b(216018N + 2L^3NR^{-1/4})$.

Altogether $|S|L \le (C_5 + 908 + 216018K_b)N + (60 + 2K_b)L^3NR^{-1/4}$, and dividing by L gives the theorem with $C_1 = 45 + C_5 + 908 + 216018K_b + 60 + 2K_b$. □

5. **Proof of Theorem 1.2**

In this section χ is a primitive character modulo $q \ge 3$.

LEMMA 5.1. *Let $B \ge 0$ be such that $\big|\sum_{m\le H} \bar\chi(m)e(m\beta)/m\big| \le B$ for every integer $H \le q/2$ and every $\beta \in \mathbb{R}$. Then $\big|\sum_{n\le x} \chi(n)\big| \le \sqrt{q}\,(B+6)$ for every $x \ge 0$.*

Proof. We may take $x = T$ to be an integer. Let $\tau = \sum_{a \bmod q} \bar\chi(a)e(a/q)$. Since $\bar\chi$ is primitive, $\sum_a \bar\chi(a)e(an/q) = \chi(n)\tau$ for every n, and summing the squares of the moduli of these identities over n mod q gives $q\varphi(q) = |\tau|^2\varphi(q)$, so $|\tau| = \sqrt{q}$. Summing over $1 \le n \le T$,

$$\tau \sum_{n\le T} \chi(n) = \sum_{0<|a|\le q/2} \bar\chi(a)\,\frac{e(aT/q) - 1}{1 - e(-a/q)},$$

where a runs over the integers with $-q/2 < a \le q/2$, $a \ne 0$. For $0 < |\theta| \le 1/2$ put $r(\theta) = (1 - e(-\theta))^{-1} - (2\pi i\theta)^{-1}$; then $|1 - e(-\theta)| = 2|\sin\pi\theta| \ge 4|\theta|$. If $2\pi|\theta| \le 1$, then $|e(-\theta) - 1 + 2\pi i\theta| \le (2\pi\theta)^2$ and $|r(\theta)| \le (2\pi\theta)^2/(4|\theta|\cdot 2\pi|\theta|) < 3$; otherwise $|r(\theta)| \le 1/(4|\theta|) + 1/(2\pi|\theta|) < 3$. The terms with $r(a/q)$ contribute at most $6q$. Writing $a = \pm m$ with $1 \le m \le q/2$, the remaining terms are $q/(2\pi i)$ times a combination, with coefficients of modulus at most 1, of four sums $\sum_{m\le H} \bar\chi(m)e(m\beta)/m$ with $H \le q/2$, so they contribute at most $2qB/\pi$. Dividing by $|\tau|$ gives the claim. □

LEMMA 5.2. *For $Y \ge 2$ and $H \ge 1$,*

$$\sum_{\substack{n\le H \\ P(n)\le Y}} \frac{1}{n} \le e^{11}\log Y.$$

Proof. The sum is at most $\prod_{p\le Y}(1-1/p)^{-1} \le \exp\bigl(\sum_{p\le Y}\frac{1}{p-1}\bigr)$, and $\sum_p\bigl(\frac{1}{p-1}-\frac{1}{p}\bigr)\le 1$; apply Lemma 2.1(iv). □

LEMMA 5.3. *There is an absolute constant C_2 such that for $0<r<q$, $b\in\mathbb{Z}$, $\beta\in\mathbb{R}$, integers $Y\ge\max(2,r)$ and $t\ge 1$,*

$$\Bigl|\sum_{\substack{n\le t\\ P(n)>Y}}\chi(n)e\Bigl(n\Bigl(\frac{b}{r}+\beta\Bigr)\Bigr)\Bigr| \le C_2\,\frac{r\log(qr+2)\,(2+2\pi|\beta|t)\,t}{Y^{1/50}\log Y}.$$

Proof. Every $n\le t$ with $P(n)>Y$ is uniquely $n=pm$ with $p=P(n)$ prime, $Y<p\le t/m$ and $P(m)\le p$, and then $m\le t/Y$. For such p we have $p>r$, so $p\nmid r$, and on the units modulo r the function $u\mapsto e(umb/r)$ equals $\sum_\psi c_\psi\psi(u)$, the sum over the characters ψ modulo r, with $|c_\psi|\le 1$.

The character $\chi\psi$ modulo qr is nonprincipal. Otherwise $\chi(n)=\bar\psi(n)$ for every n prime to qr. Let $d=(q,r)$ and let n be prime to q with $n\equiv 1 \pmod d$. By the Chinese remainder theorem there is n' prime to qr with $n'\equiv n \pmod q$ and $n'\equiv 1 \pmod r$, and then $\chi(n)=\chi(n')=\bar\psi(n')=1$. So χ is trivial on the units congruent to 1 modulo d, and is induced by a character modulo $d\le r<q$, which contradicts primitivity.

Applying (3.1) to $\xi=\chi\psi$ with $\gamma=m\beta$, $A=\max(Y,P(m)-1)$ and $B=\lfloor t/m\rfloor$ (the inner sum is empty if $A\ge B$) gives

$$\Bigl|\sum_{\substack{n\le t\\ P(n)>Y}}\chi(n)e\Bigl(n\Bigl(\frac{b}{r}+\beta\Bigr)\Bigr)\Bigr| \le 2C_3 r\,\frac{\log(qr+2)(2+2\pi|\beta|t)}{\log Y}\sum_{m\le t/Y}\Bigl(\frac{t}{m}\Bigr)^{49/50},$$

and $\sum_{m\le M} m^{-49/50}\le 50M^{1/50}$. Hence $C_2=100C_3$ will do. □

PROPOSITION 5.4. *There is an absolute constant C_4 such that if $\log q\ge 200$, then $\bigl|\sum_{m\le H}\chi(m)e(m\beta)/m\bigr|\le C_4\log\log q$ for every integer $H\le q$ and every $\beta\in\mathbb{R}$.*

Proof. Let $L=\log q$, $M=\lceil L\rceil$, $R=M^{16}$ and $Y=M^{1800}$. Since $e^{L/16}\ge (L/16)^4/24 > L+1\ge M$, we have $2\le R<q$ and $R\le Y$. Split the sum according to $P(m)\le Y$ or $P(m)>Y$. By Lemma 5.2 the first part is at most $e^{11}\log Y$. For the second put $A(t)=\sum_{n\le t,\,P(n)>Y}\chi(n)e(n\beta)$ and $w(t)=1/t-1/(t+1)$. Partial summation and $|A(H)|\le H$ give

$$\Bigl|\sum_{\substack{m\le H\\ P(m)>Y}}\frac{\chi(m)e(m\beta)}{m}\Bigr| \le 1+\sum_{1\le t<H}|A(t)|\,w(t).$$

If $t\le Y$, then $A(t)=0$. Let $t>Y$ and $Q=\lfloor t/R\rfloor\ge 1$. By Dirichlet's theorem there are coprime a,r with $1\le r\le Q$ and $|\beta-a/r|\le 1/(r(Q+1))$; then $|\beta-a/r|\,t\le R$ (Figure 3). If $r\le R$, then Lemma 5.3 with $b=a$ gives

$|A(t)| \le C_2 k t$, where

$$k = \frac{R\log(qR+2)(2+2\pi R)}{Y^{1/50}\log Y}.$$

If $r > R$, then $R \le r \le t/R$ and $|\beta - a/r| \le r^{-2}$, so Theorem 4.1 for $f = \chi$ bounds the full sum $\sum_{n\le t}\chi(n)e(n\beta)$, and since $\log t \le L$, removing the $\Psi(t) = \#\{n \le t : P(n) \le Y\}$ smooth terms gives

$$|A(t)| \le C_1\Big(\frac{t}{\log t} + t\rho\Big) + \Psi(t), \qquad \rho = \frac{L^2}{R^{1/4}} = \frac{L^2}{M^4}.$$

Since $tw(t) \le 1/t$ and $w(t) \le 1/t^2$,

$$\sum_{1\le t<H} |A(t)|\,w(t) \le C_1 \sum_{2\le t\le H} \frac{1}{t\log t} + (C_1\rho + C_2 k)\sum_{t\le H}\frac{1}{t} + \sum_{1\le t<H}\Psi(t)w(t).$$

The first sum is at most $3+\log L$, the second at most $1+L$, and by partial summation the third is at most $\sum_{n\le H,\,P(n)\le Y} 1/n \le e^{11}\log Y$. Finally $L \le M \le 2L$ gives $\rho(1+L) \le 2/M \le 1$; $\log(qR+2) \le 18M$, $2+2\pi R \le 8M^{16}$, $Y^{1/50} = M^{36}$ and $\log Y = 1800\log M \ge 1800$ give $k(1+L) \le 288M^{34}/(1800M^{36}) \le 1$; and $\log Y = 1800\log M \le 3600\log L$. Collecting terms, the sum is at most $1 + 4C_1 + C_2 + (7200e^{11} + C_1)\log L \le (7200e^{11} + 1 + 5C_1 + C_2)\log L$. □

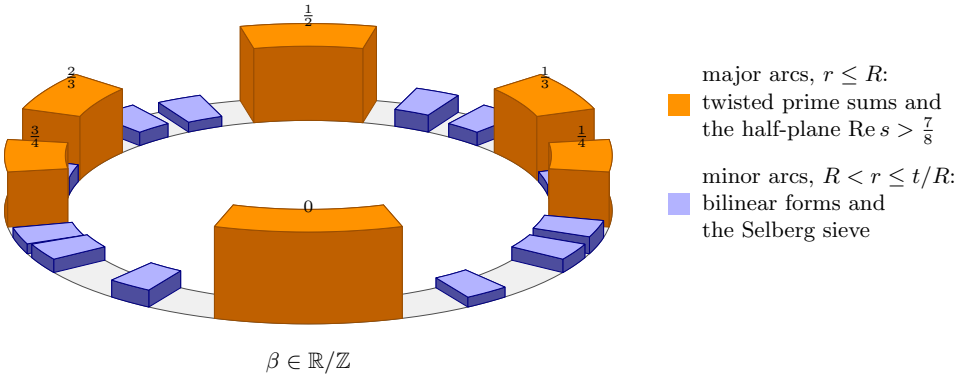


Figure 3. The dissection of $\mathbb{R}/\mathbb{Z}$ in the proof of Proposition 5.4, drawn for $R = 4$. Near a fraction a/r with $r \le R$ (major arcs) the sum $A(t)$ is controlled by the twisted prime sums of Corollary 3.8; elsewhere (minor arcs) by Theorem 4.1. The heights only indicate that the major arcs carry the large values.

Proof of Theorem 1.2. The character $\bar{\chi}$ is primitive modulo q. If $\log q \geq 200$, then Proposition 5.4 for $\bar{\chi}$ and Lemma 5.1 give

$$\Big|\sum_{n\leq x}\chi(n)\Big| \leq \sqrt{q}\,(C_4\log\log q + 6) \leq (C_4+6)\sqrt{q}\log\log q.$$

If $\log q < 200$, then every sum $\sum_{m\leq H}\bar{\chi}(m)e(m\beta)/m$ with $H \leq q$ has modulus at most $1+\log q < 201$, and Lemma 5.1 gives

$$\Big|\sum_{n\leq x}\chi(n)\Big| \leq 207\sqrt{q} \leq \frac{207}{\log\log 3}\sqrt{q}\log\log q. \qquad \square$$

6. The logarithmic derivative at $s = 1$

Recall that $\sigma_0 = 11/12 + 3/1000$ and that K is the constant of Corollary 3.8.

LEMMA 6.1. *For every nonprincipal character χ modulo q,*

$$-\frac{L'}{L}(1,\chi) = 2\int_1^\infty \frac{S(\chi;t)}{t^2}\,dt,$$

and the integral converges absolutely.

Proof. By Lemma 3.1(i) with $a = \chi\Lambda$, for real $s > 1$,

$$\int_1^\infty S(\chi;t)\,t^{-s-1}\,dt = -\frac{L'}{L}(s,\chi)\,\frac{1}{s(s+1)}.$$

As $s \to 1^+$ the left side tends to $\int_1^\infty S(\chi;t)t^{-2}dt$ by dominated convergence, since $|S(\chi;t)| \leq Kt^{\sigma_0}\log(q+2)$ and $\sigma_0 < 1$. The function $L(s,\chi)$ is entire and does not vanish at $s = 1$ by Theorem 1.1, so the right side tends to $-\frac{1}{2}\frac{L'}{L}(1,\chi)$. $\square$

Proof of Theorem 1.3(i). Let χ be nonprincipal modulo q, so $q \geq 3$, and put $Y = \big(\log(q+2)\big)^{1/(1-\sigma_0)} \geq 1$. Since $|S(\chi;t)| \leq \psi(t) \leq 4t$ by Lemma 2.1(ii), Lemma 6.1 gives

$$\Big|\frac{L'}{L}(1,\chi)\Big| \leq 2\int_1^Y \frac{4\,dt}{t} + 2K\log(q+2)\int_Y^\infty t^{\sigma_0-2}\,dt = \frac{8\log\log(q+2)+2K}{1-\sigma_0}.$$

As $q+2 \leq q^2$, $\log\log(q+2) \leq \log 2 + \log\log q$, and since $\log\log q \geq \log\log 3 > 0$ the right side is at most $C\log\log q$. $\square$

For the lower bound in Theorem 1.3(ii) we need to control the prime powers $n \leq q^{25}$ with $n \equiv 1 \pmod q$. We count them in dyadic blocks with Selberg's sieve.

LEMMA 6.2. *Let q be prime, let $M \geq 1$ be an integer and let $1 < z < q$. The number of m with $M \leq m < 2M$ for which $1 + mq$ is a prime power is at most*

$$\frac{2M}{\log z} + z(1 + \log z)^2 + z.$$

Proof. If $1 + mq = p^k$ with $p \leq z$, then $p^k \in (Mq, 2Mq)$, and for each prime p at most one power of p lies in this interval, since $p^{k+1} \geq 2p^k$. These m number at most z. If $p > z$, then $1 + mq$ has no prime factor up to z, and we apply Lemma 2.2 to the M integers $a_m = 1 + mq$, $M \leq m < 2M$, with P the product of the primes up to z, $\nu(d) = 1/d$ and $y = z$. Every prime $p \leq z$ differs from q, so for $d \mid P$ the condition $d \mid 1 + mq$ fixes m in one residue class modulo d, and $|R_d| \leq 1$. Here $g(l) = \prod_{p \mid l}(p-1)^{-1}$, which is the sum of $1/n$ over the n whose set of prime factors is that of l. Every $n \leq \sqrt{z}$ has all its prime factors at most z and the product of its distinct prime factors is at most $\sqrt{z}$, so

$$S = \sum_{\substack{l \mid P \\ l \leq \sqrt{z}}} g(l) \geq \sum_{n \leq \sqrt{z}} \frac{1}{n} \geq \frac{\log z}{2}.$$

For squarefree d, $3^{\omega(d)} = \tau_3(d)$, and $\sum_{d \leq z} \tau_3(d) = \#\{(a,b,c) : abc \leq z\} \leq z\big(\sum_{a \leq z} 1/a\big)^2 \leq z(1 + \log z)^2$. Lemma 2.2 therefore bounds the number of these m by $2M/\log z + z(1 + \log z)^2$. □

LEMMA 6.3. *There is an absolute constant C_0 such that for every prime $q \geq 3$,*

$$\sum_{\substack{1 < n \leq q^{25} \\ n \equiv 1 \pmod q}} \frac{\Lambda(n)}{n} \leq C_0 \frac{\log q \log\log q}{q}.$$

Proof. Write $n = 1 + mq$ with $1 \leq m < q^{24}$, and let $J = \lfloor 24 \log_2 q \rfloor + 1$, so that $2^J > q^{24}$ and $J \leq 36 \log q$. We split the range of m into the blocks $2^j \leq m < 2^{j+1}$, $0 \leq j < J$ (Figure 4), and let N_j be the number of m in the j-th block with $1 + mq$ a prime power. We claim $N_j \leq 600 \cdot 2^j/(j+1)$. For $j < 48$ this holds trivially. For $j \geq 48$ apply Lemma 6.2 with $M = 2^j$ and $z = 2^u$, $u = j/48 \geq 1$. Since $2^j < 2^J \leq 2q^{24}$, we have $z < 2^{1/48} q^{1/2} < q$. Then $2M/\log z = 96 \cdot 2^j/(j \log 2) \leq 150 \cdot 2^j/(j+1)$, and since $\log z \leq u$,

$$z(1 + \log z)^2 + z \leq 2^u (2 + u)^2 \leq \frac{441 \cdot 2^{4u}}{48u + 1} \leq \frac{441 \cdot 2^j}{j+1},$$

because $(2+u)^2(48u+1) \leq 9u^2 \cdot 49u = 441u^3$ and $u^3 \leq 2^{3u}$. This proves the claim.

A term of the j-th block has $\Lambda(n) \le \log(1+2^{j+1}q) \le (j+2)\log 2 + \log q$ and $n > 2^j q$, so the block contributes at most

$$\frac{600}{(j+1)q}\big((j+2)\log 2 + \log q\big) \le \frac{600}{q}\Big(2\log 2 + \frac{\log q}{j+1}\Big).$$

Summing over $j < J$ and using $\sum_{j<J} 1/(j+1) \le 1 + \log J$, $J \le 36\log q$ and $\log J \le 4 + \log\log q$, the sum in the lemma is at most $\frac{600}{q}\big(51\log q + 5\log q + \log q \log\log q\big)$. Since $\log\log q \ge \log\log 3 > 1/11$, this is at most $C_0 \log q \log\log q/q$ with $C_0 = 600 \cdot 617$. □

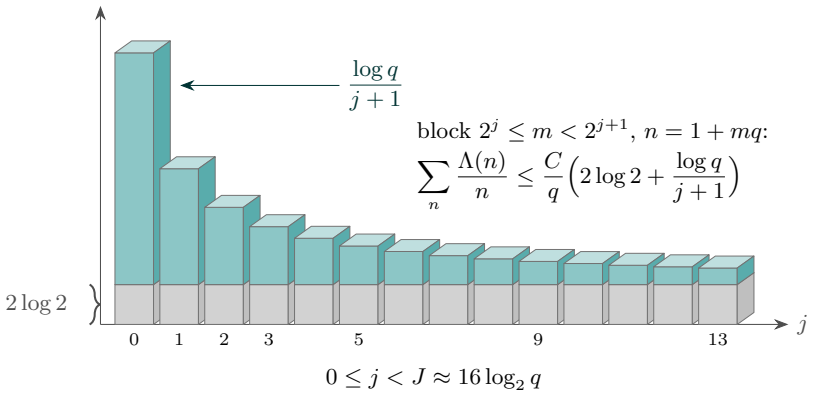


Figure 4. The dyadic blocks in the proof of Lemma 6.3. The block $2^j \le m < 2^{j+1}$ contributes a constant part, of total size $\ll J \ll \log q$, and a part of size $\log q/(j+1)$, whose total is $\ll \log q \log J$. The second part produces the factor $\log\log q$.

Proof of Theorem 1.3(ii). Let $q \ge 3$ be prime and put $c(n) = \sum_{\chi \ne \chi_0} \chi(n)$, the sum over the nonprincipal characters modulo q. By orthogonality $c(n) = (q-1)\mathbf{1}[n \equiv 1 \pmod q] - \mathbf{1}[q \nmid n]$. Put

$$T(t) = \sum_{\chi \ne \chi_0} S(\chi; t) = \sum_{n \le t} c(n)\Lambda(n)\Big(1 - \frac{n}{t}\Big), \qquad I = \int_1^\infty \frac{T(t)}{t^2}\,dt.$$

By Lemma 6.1, $\sum_{\chi \ne \chi_0} \frac{L'}{L}(1,\chi) = -2I$, so by (1.1) it suffices to show $-C\log q \le I \le C\log q \log\log q$.

By Corollary 3.8(i), $|T(t)| \le qK\log(q+2)\, t^{\sigma_0}$. Let $Y = q^{25}$. Since $25(\sigma_0 - 1) < -2$ and $\log(q+2) \le q$,

$$\Big|\int_Y^\infty \frac{T(t)}{t^2}\,dt\Big| \le \frac{qK\log(q+2)\, Y^{\sigma_0 - 1}}{1-\sigma_0} \le \frac{K\log(q+2)}{q(1-\sigma_0)} \le \frac{K}{1-\sigma_0}.$$

For the lower bound, $c(n) \geq -1$ gives $-T(t) \leq \psi(t) \leq 4t$, so

$$-I \leq \int_1^Y \frac{4\,dt}{t} + \frac{K}{1-\sigma_0} = 100\log q + \frac{K}{1-\sigma_0}.$$

For the upper bound, $c(n) \leq (q-1)\mathbf{1}[n \equiv 1 \pmod q]$ gives

$$\begin{aligned}\int_1^Y \frac{T(t)}{t^2}\,dt &\leq (q-1) \sum_{\substack{n\leq Y \\ n\equiv 1 \pmod q}} \Lambda(n) \int_n^\infty \Big(1-\frac{n}{t}\Big)\frac{dt}{t^2} \\ &= \frac{q-1}{2} \sum_{\substack{1<n\leq q^{25} \\ n\equiv 1 \pmod q}} \frac{\Lambda(n)}{n},\end{aligned}$$

which is at most $\frac{C_0}{2}\log q \log\log q$ by Lemma 6.3. Hence

$$I \leq \frac{C_0}{2}\log q\log\log q + \frac{K}{1-\sigma_0},$$

and both bounds follow since $\log q > 1$ and $\log\log q > 1/11$. □

Appendix A. Transfer parameters for the eleven-twelfths half-plane

The purpose of this appendix is to check the numerical consequences of changing the zero-free boundary. The arithmetic here does not prove the zero-free theorem. It checks that the subsequent deductions use only the cited zero-free input. No claim of novelty is made for these elementary parameter choices.

A.1. *The contour and unsmoothing margins.* Put

$$\theta = \frac{11}{12}, \qquad \delta = \frac{1}{2000}, \qquad a = \theta - 2\delta, \qquad \sigma_0 = a + 8\delta = \frac{2759}{3000}.$$

We have $1/2 < a < 1$, $0 < \delta < 1/1000$, and $a + 2\delta = \theta$. Thus the hypothesis of Proposition 3.7 follows from the cited nonvanishing statement. That proposition gives $|S(\chi;x)| \leq Kx^{\sigma_0}\log(q+2)$.

In Corollary 3.8 choose $h = x^{47/50}$. The two exponents arising from unsmoothing satisfy

$$\frac{49}{50} - \Big(1 + \sigma_0 - \frac{47}{50}\Big) = \frac{1}{3000} > 0, \qquad \frac{47}{50} + \frac{1}{100} = \frac{19}{20} < \frac{49}{50}.$$

The first inequality controls the difference of two Riesz means divided by h; the second absorbs the logarithm in the short-interval error. The prime-power correction is smaller still, since $1/2+1/100 < 49/50$. Consequently the proof gives a twisted prime-sum bound with exponent $49/50$, with constants depending only on the fixed choice of δ.

A.2. *Major arcs.* Partial summation gives, for $M \geq 1$,

$$\sum_{m \leq M} m^{-49/50} \leq 1 + \int_1^M u^{-49/50}\, du = 50M^{1/50} - 49 \leq 50M^{1/50}.$$

The proof of Lemma 5.3 therefore gives its stated bound with $Y^{1/50}$ in the denominator and $C_2 = 100C_3$.

Let $L = \log q \geq 200$ and $M = \lceil L \rceil$, as in Proposition 5.4. Set $R = M^{16}$ and $Y = M^{1800}$. The elementary inequalities $L \leq M \leq 2L$ and $e^{L/16} \geq (L/16)^4/24 > L + 1$ show $2 \leq R < q$. For the last strict inequality it suffices that $L^3 > 48 \cdot 16^4$, which holds for $L \geq 200$.

In the major-arc contribution, $\log(qR + 2) \leq 18M$, $2 + 2\pi R \leq 8M^{16}$ and $1 + L \leq 2M$. Hence

$$k(1 + L) \leq \frac{288M^{34}}{1800M^{36} \log M} \leq 1.$$

Here $1800/50 = 36$, leaving two powers of M beyond the required numerator exponent 34. The minor-arc estimate is unchanged: $\rho(1 + L) \leq 2/M \leq 1$. Finally $\log Y = 1800 \log M \leq 3600 \log L$, so the two smooth-number terms contribute at most $7200e^{11} \log L$. These are exactly the constants used in Proposition 5.4.

The old choice $Y = M^{700}$ would give $Y^{1/50} = M^{14}$, which does not establish the displayed major-arc bound. Changing the boundary without changing this parameter would leave an actual gap.

A.3. *The cyclotomic tail and progression sieve.* For the tail in Theorem 1.3(ii), choose $Y = q^{25}$. The exact margin is

$$25(1 - \sigma_0) - 2 = \frac{1}{120} > 0.$$

Thus $q \log(q + 2) Y^{\sigma_0 - 1} \leq \log(q + 2)/q \leq 1$ for $q \geq 3$. The corresponding old margin, with 17 in place of 25, is negative; the old tail estimate cannot simply be retained.

Now $n = 1 + mq \leq q^{25}$ implies $m < q^{24}$. Put $J = \lfloor 24 \log_2 q \rfloor + 1$. Since $\log 2 > 56/81$ and $\log q > 1$, we have

$$J \leq \left(\frac{24 \cdot 81}{56} + 1 \right) \log q < 36 \log q.$$

The bound for $\log 2$ follows from the first two terms of

$$\log 2 = 2 \sum_{j \geq 0} \frac{(1/3)^{2j+1}}{2j + 1}.$$

For $j \geq 48$ use $u = j/48$ and $z = 2^u$ in Lemma 6.2. Then $1 < z < 2^{1/48}q^{1/2} < q$, and

$$\frac{2 \cdot 2^j}{\log z} = \frac{96 \cdot 2^j}{j \log 2} \leq \frac{150 \cdot 2^j}{j+1}.$$

Indeed, $\log 2 > 2/3$ and $(j+1)/j \leq 49/48$ make the ratio at most $96(3/2)(49/48) = 147 < 150$. For the remaining term,

$$(2+u)^2(48u+1) \leq 9u^2 \cdot 49u = 441u^3 \leq 441\, 2^{3u} \quad (u \geq 1),$$

and therefore

$$2^u(2+u)^2 \leq \frac{441 \cdot 2^{4u}}{48u+1} \leq \frac{441 \cdot 2^j}{j+1}.$$

The elementary inequality $u \leq 2^u$ for $u \geq 1$ follows, for example, by minimizing $u \log 2 - \log u$. We obtain $N_j \leq 591 \cdot 2^j/(j+1) \leq 600 \cdot 2^j/(j+1)$. For $j < 48$ the bound with 600 is immediate from $N_j \leq 2^j$.

Summing the blocks as in Lemma 6.3 gives

$$\sum_{\substack{1<n\leq q^{25} \\ n\equiv 1 \pmod q}} \frac{\Lambda(n)}{n} \leq \frac{600}{q}\big(56 \log q + \log q \log\log q\big).$$

Here $2(\log 2)J \leq 72(\log 2)\log q < 51 \log q$ and $1 + \log J \leq 5 + \log\log q$. Since $\log\log 3 > 1/11$, the constant $C_0 = 600 \cdot 617$ suffices. The lower-bound integral now contributes $4\log(q^{25}) = 100 \log q$ in place of $68 \log q$. This completes the parameter verification used in the article.

A.4. *What was formally checked.* The three rational margin statements, including the two failures of the old cutoffs, were proved using `norm_num` in the accompanying Lean file `ParameterAudit.lean`. Fresh axiom output lists only `propext`, `Classical.choice` and `Quot.sound`. The file does not claim to formalise all analytic estimates in this appendix or to prove the cited zero-free theorem. The exact rational Python program is a separate consistency check, with the same limits.

Appendix B. Uniform transfer from a fixed zero-free boundary

The role of the zero-free theorem can be separated from the numerical choices of Appendix A. The following is a conditional transfer statement; its proof supplies no independent zero-free theorem.

PROPOSITION B.1. *Fix $1/2 \leq \theta < 1$. Suppose that every nonprincipal Dirichlet character ξ has $L(s,\xi) \neq 0$ for $\operatorname{Re} s > \theta$. There exists $C_\theta > 0$ such that every primitive character χ modulo $q \geq 3$ satisfies*

$$\sup_{x \geq 0} \left| \sum_{n \leq x} \chi(n) \right| \leq C_\theta \sqrt{q} \log\log q.$$

The constant is uniform in the modulus, the character and the summation endpoint. Its dependence on θ is allowed.

Proof. Choose $\delta = \min(1/2000, (1-\theta)/100)$ and put $a = \max(1/2, \theta-2\delta)$, $\sigma = a+8\delta$ and $\eta = (1-\sigma)/4$. If $a = \theta-2\delta$, then $\sigma = \theta+6\delta \le \theta+6(1-\theta)/100 < 1$. If $a = 1/2$, then $\sigma \le 1/2 + 8/2000 < 1$. Thus $1/2 \le a \le 1$, $0 < \delta < 1/1000$, $0 < \eta \le 1/8$ and $a + 2\delta \ge \theta$. Proposition 3.7 therefore gives, uniformly for nonprincipal ξ modulo k,

$$|S(\xi; x)| \ll_\theta x^\sigma \log(k+2).$$

The unsmoothing identity in Corollary 3.8, with $h = x^{(1+\sigma)/2} = x^{1-2\eta}$, gives

$$\left|\sum_{p \le x} \xi(p) \log p\right| \ll_\theta x^{1-\eta} \log(k+2).$$

Here the difference of the two Riesz means divided by h has exponent $1-2\eta$, the interval error is $O(x^{1-2\eta}\log(2x))$, and the prime-power error is $O(\sqrt{x}\log x)$. The inequality $\log x \le x^\eta/\eta$ absorbs both logarithms; $\sigma \ge 1/2$ ensures $1/2 + \eta \le 1-\eta$. Partial summation gives the corresponding additively twisted prime bound.

The proof of Lemma 5.3 now uses

$$\sum_{m \le M} m^{-1+\eta} \le M^\eta/\eta$$

and gives the same conclusion with Y^η in the denominator, with an implied constant depending on θ. For $L = \log q \ge 200$ and $M = \lceil L \rceil$, take $R = M^{16}$ and $Y = M^A$, where A is an integer with $A\eta \ge 36$ and $A \ge 16$. The minor-arc theorem is unchanged. Its weighted contribution is bounded by $O(1 + \log L)$, since $\rho(1+L) \le 1$. For the major arcs the coefficient after Abel summation is bounded by

$$\frac{R\log(qR+2)(2+2\pi R)(1+L)}{Y^\eta \log Y} \le \frac{288 M^{34-A\eta}}{A \log M} \le 288.$$

Finally, the two smooth-number contributions have total at most $2e^{11}\log Y \le 4Ae^{11}\log L$. This proves the harmonic-sum bound $O_\theta(\log\log q)$ exactly as in Proposition 5.4, including the case $Y \ge H$, in which the nonsmooth sum is empty. Lemma 5.1 proves the result for $L \ge 200$. For $L < 200$ its elementary harmonic bound gives $207\sqrt{q}$, absorbed using $\log\log q \ge \log\log 3 > 0$. □

This proposition proves the transfer from a fixed zero-free boundary to the stated character-sum bound. It does not prove the existence of such a boundary. Theorem 1.1 is the cited external result that supplies that antecedent in the main application.

Appendix C. Smoothed twisted Möbius cancellation

This appendix makes the dependence of a smoothed Möbius estimate on the zero-free input precise. The forward implication below is a consequence of nonvanishing, not an independent proof of it. We allow complex-valued weights. For $W \in C_c^\infty((1,2);\mathbb{C})$ put

$$S_{\chi,W}(X) = \sum_{n\ge 1} \mu(n)\chi(n)W(n/X), \qquad \widehat{W}(s) = \int_0^\infty W(y)y^{s-1}\,dy.$$

The sum is finite for every $X > 0$.

Lemma C.1 (Polynomial growth of the reciprocal). *Let $0 < \theta < 1$, let χ be a nonprincipal character modulo q, and suppose $L(s,\chi) \neq 0$ on $\operatorname{Re} s > \theta$. If $0 < d < 1-\theta$, there are constants $B > 0$ and $C > 0$, depending only on θ, d, such that*

$$|L(u+it,\chi)^{-1}| \le Cq^B(1+|t|)^B \quad (\theta + d \le u \le 2).$$

For $\theta = 7/8$ and $d = 1/24$ one may take the explicit bound

$$|L(u+it,\chi)^{-1}| \le 2^{105}\big(7q(1+|t|)\big)^{104} \quad (11/12 \le u \le 2). \tag{C.1}$$

Proof. Since χ is nonprincipal, the sum over a full period vanishes. Thus $A(v) = \sum_{n\le v}\chi(n)$ satisfies $|A(v)| \le q$. Partial summation gives

$$L(s,\chi) = s\int_1^\infty A(v)v^{-s-1}\,dv, \qquad |L(s,\chi)| \le \frac{q|s|}{\operatorname{Re} s} \quad (\operatorname{Re} s > 0).$$

The integral is locally uniformly convergent, together with its derivatives, on that half-plane. It therefore gives the needed holomorphic continuation.

Fix $t \in \mathbb{R}$ and centre a disc at $c = 2 + it$. Set

$$R = 2 - \theta - d/2, \quad r = 2 - \theta - d, \quad U = \log\left(2 + \frac{q(|t|+4)}{\theta + d/2}\right).$$

The closed disc of radius R lies inside the assumed zero-free half-plane. There is a holomorphic branch $g = \log L$ on a neighbourhood of this disc, normalised at c by the absolutely convergent Euler-product logarithm. Consequently $|g(c)| \le \log\zeta(2) < \log 2$, whereas $\operatorname{Re} g < U$ throughout the disc. The positive harmonic function $U - \operatorname{Re} g$ obeys Harnack's inequality: for $|s-c| \le r$,

$$U - \operatorname{Re} g(s) \le K\big(U - \operatorname{Re} g(c)\big), \qquad K = \frac{R+r}{R-r}.$$

For completeness, this inequality follows from the Poisson integral on a concentric circle of radius $R' < R$: its positive kernel at radius at most r is at most $(R'+r)/(R'-r)$ times the kernel at the centre. Letting $R' \uparrow R$ gives the stated factor. It follows that

$$|L(s,\chi)^{-1}| \le 2^K e^{(K-1)U}.$$

Every $u+it$ in the stated segment is in the smaller disc. Since $K>1$, this proves the first assertion with $B=K-1$. For the specified numerical values, $R=53/48$, $r=52/48$, $K=105$, and $\theta+d/2=43/48$. Finally

$$2+\frac{48q(|t|+4)}{43}\le 7q(1+|t|)$$

for $q\ge 1$. This proves (C.1). □

Proposition C.2 (The zero-free and smoothed formulations). *Fix a non-principal Dirichlet character χ and $0<\theta<1$. The following statements are equivalent:*

(i) $L(s,\chi)\neq 0$ *whenever* $\operatorname{Re} s>\theta$.

(ii) *For every $\varepsilon>0$ and every $W\in C_c^\infty((1,2);\mathbb{C})$ there is a constant $C_{\chi,W,\varepsilon}$ such that, for all $X\ge 1$,*

$$|S_{\chi,W}(X)|\le C_{\chi,W,\varepsilon}X^{\theta+\varepsilon}.$$

Proof. Assume (i). It suffices to consider $0<\varepsilon<1-\theta$; larger ε follow from the trivial bound $|S_{\chi,W}(X)|\le 3\|W\|_\infty X$ for $X\ge 1$. The Euler product, absolute convergence and Mellin inversion give

$$S_{\chi,W}(X)=\frac{1}{2\pi i}\int_{2-i\infty}^{2+i\infty}\frac{\widehat{W}(s)X^s}{L(s,\chi)}\,ds. \tag{C.2}$$

Indeed, $L(s,\chi)^{-1}=\sum_{n\ge1}\mu(n)\chi(n)n^{-s}$ on $\operatorname{Re} s>1$; the series and integral can be interchanged absolutely on $\operatorname{Re} s=2$. The substitution $y=e^v$ and repeated integration by parts show that for every integer $J\ge 0$,

$$|\widehat{W}(u+it)|\le C_{W,J,\theta,\varepsilon}(1+|t|)^{-J}\quad(\theta+\varepsilon\le u\le 2).$$

The boundary terms vanish because W has compact support inside $(1,2)$. Lemma C.1, with $d=\varepsilon$, supplies a polynomial bound for the reciprocal throughout this strip. Choose $J>B+2$. There are no poles in the strip, the horizontal sides of the contour tend to zero as their heights tend to infinity, and both vertical integrals converge absolutely. Shifting (C.2) to $\operatorname{Re} s=\theta+\varepsilon$ proves (ii). The constants in this implication may be bounded by a constant depending on W,θ,ε times q^B.

Conversely assume (ii). For each fixed W the function

$$F_W(s)=\int_0^\infty S_{\chi,W}(X)X^{-s-1}\,dX$$

is holomorphic on $\operatorname{Re} s>\theta$. Near zero the integrand vanishes for $X\le 1/2$; near infinity, choose ε smaller than the distance of a compact subset from $\operatorname{Re} s=\theta$. This gives locally uniform convergence of the integral and its derivatives. For

$\operatorname{Re} s > 1$, absolute convergence and the substitution $y = n/X$ give

$$F_W(s) = \frac{\widehat{W}(s)}{L(s,\chi)}.$$

Since $L(s,\chi)$ is holomorphic on $\operatorname{Re} s > 0$, the identity theorem gives $L(s,\chi)F_W(s) = \widehat{W}(s)$ on $\operatorname{Re} s > \theta$. If ρ were a zero in this region, choose a nonzero nonnegative $\varphi \in C_c^\infty((1,2);\mathbb{R})$ and set $W(y) = y^{-\rho}\varphi(y)$. This is an admissible complex weight, with

$$\widehat{W}(\rho) = \int_1^2 \frac{\varphi(y)}{y}\,dy > 0.$$

Evaluating the continued identity at ρ gives a contradiction. □

Corollary C.3 (Möbius consequence of the credited input). *Using OpenAI's 7/8 theorem cited in Theorem 1.1, for every $\varepsilon > 0$ and fixed nonprincipal χ,*

$$S_{\chi,W}(X) \ll_{\chi,W,\varepsilon} X^{7/8+\varepsilon}.$$

In particular, for every $W \in C_c^\infty((1,2);\mathbb{C})$ there is a constant C_W such that, uniformly in nonprincipal characters modulo q,

$$|S_{\chi,W}(X)| \le C_W q^{104} X^{11/12} \qquad (X \ge 1). \tag{C.3}$$

Proof. The first assertion is Proposition C.2 with $\theta = 7/8$. For the second, shift (C.2) to $11/12$ and use (C.1). The finite quantity

$$A_W = \sup_{\substack{11/12 \le u \le 2 \\ t \in \mathbb{R}}} (1+|t|)^{107} |\widehat{W}(u+it)|$$

gives the choice $C_W = 2^{105}7^{104}A_W/(2\pi)$, since $\int_{-\infty}^{\infty}(1+|t|)^{-3}\,dt = 1$. □

The corollary proves the requested smoothed estimate from the cited theorem. Proposition C.2 also shows precisely why using that estimate to reprove the same nonvanishing would not remove the dependency. No independent proof of assertion (i) or (ii) is supplied here. These are written analytic deductions; they are not asserted to be new Lean declarations or a new zero-free theorem.

Source and computational material

The source archive [2] and the public preprint https://hexagonmath.org/2610.00256v2 provide the earlier coauthored versions. Sources for this revision, vector figures, numerical checks and separate verification notes are available in the same repository. The parameter checks verify stated arithmetic margins; they do not establish Theorem 1.1.

Use of AI tools

The earlier coauthored manuscript and Lean development used Claude under the authors' direction, as recorded in the source archive. OpenAI Codex prepared the explicit parameter conversion in Section 3, Lemma 5.3, Proposition 5.4, and the progression and tail estimates in Section 6. It expanded the Mellin inversion justification and supplied Appendices B and C. Appendix A identifies the exact numerical margins. The authors remain responsible for the mathematics and references.

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